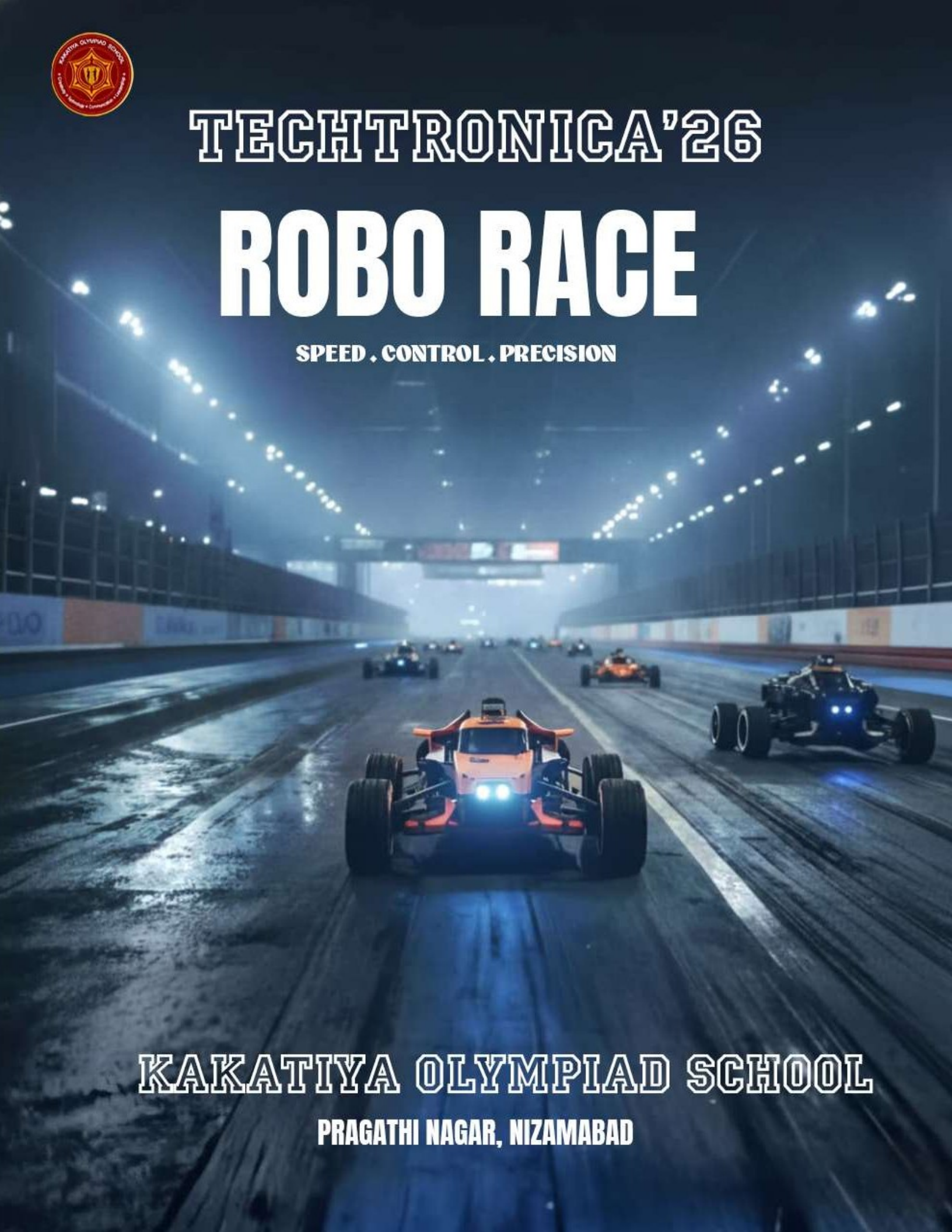
TECHTRONICA'26

ROBO RACE

SPEED . CONTROL . PRECISION



KAKATIYA OLYMPIAD SCHOOL

PRAGATHI NAGAR, NIZAMABAD

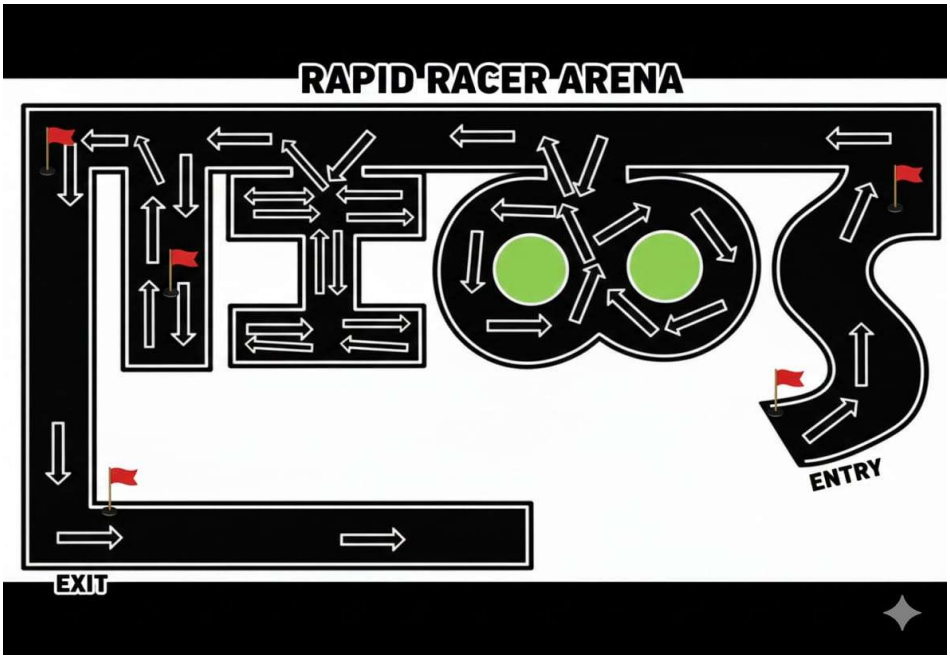
ROBO RACE

1. THE CHALLENGE

Objective:

Design a wired or wireless robot to navigate a complex race course. The robot that completes the track with the fastest time (including penalties) wins!

2. THE ARENA & EQUIPMENT



- **Track Width:** Minimum of 30cm wide.
- **Surface:** The arena is ground-level and contains various turns and obstacles.
- **Design:** The track layout is designed to test both speed and steering precision.

3. TEAM REQUIREMENTS

- **Eligibility:** Open to students in Grades 7, 8, and 9 (and up to PG as per original rules).
- **Composition:** Teams can be made of students from different sections or institutions.

- **Arena Access:** Only **two members** per team are allowed at the arena side (one Pilot, one Technician).
- **Single Entry:** A team plays with **one robot**. You may register up to 3 times only if the robot is significantly different (different chassis, motors, or wheels).

4. ROBOT SPECIFICATIONS

To keep the race fair, all robots must meet these "Pre-Match Inspection" standards:

Feature	Requirement
Size	Max 30cm x 30cm x 30cm (5% tolerance allowed)
Track Width	Minimum wheel-to-wheel width of 25cm x 25cm
Weight	Maximum 3kg (Batteries count toward weight if on-board)
Power	Maximum 12V DC (5% tolerance allowed); **Battery only**
Control	Wired or Wireless (Wired cables must be at least 3m long)
Forbidden Items	No Lego kits, no IC (gas) engines, no arena-damaging parts

5. THE COMPETITION STRUCTURE

Round 1: The Qualifier

- **Format:** Solo time-trial (one team at a time).
- **Goal:** Complete 1 full lap of the circuit within the **5-minute** time limit.
- **Penalties:** Touching any barricade will result in a time penalty (added seconds).
- **Advancement:** The top 12 fastest teams (Time + Penalties) move to the Final Round.

The Final Round

- **Format:** 1 lap circuit race.
- **Winner:** The team that completes the arena in the least amount of total time.
- **Requirement:** Must use the same robot from the Qualifiers. Repairs are only allowed if replacement parts are identical to the originals

6. GAMEPLAY RULES & FOULS

- **Technical Timeout:** You get **two 30-second repair windows**. After these are used, the timer will not stop for any reason.
- **Power Supply:** Batteries can be on or off-board (held by a team member), but cannot be changed during the race. No external plug-in power is allowed.
- **Movement:** The robot must not expand or compress during the run.
- **Human Intervention:** If a participant touches the robot more than **2 times** (outside of a technical timeout), the team is disqualified.

7. DISQUALIFICATIONS & SAFETY

Your team will be disqualified for:

- **Damaging the Arena:** Any robot that marks or breaks the track is out.
- **Safety:** Participants must bring all their own tools and accessories.
- **Illegal Sharing:** Lending your robot to another team is strictly forbidden.
- **Bad Sportsmanship:** Misbehaviour toward coordinators, mentors, or other teams.

Note: The coordinators' decisions are final. Always check with your robotics mentors for any last-minute updates on the official website.